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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 03/29/2006 |
| **PI Name:** | Liang, Shoudan Ph.D. | | |
| **Project Title:** | High-Throughput Metabolic Profiling by Multidimensional NMR and Mathematical Modeling of Metabolic Networks | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | BIOTECHNOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | BIOTECHNOLOGY--Biomolecular physics & chemistry | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | INTERDISCIPLINARY (2004-2007) | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 01/01/2002 | **End Date:** | 09/30/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA HQ |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Liang, Shoudan | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Recently-developed high-throughput techniques such as DNA micro-arrays have made it possible to simultaneously monitor the expression levels of all the genes in a cell. However, elucidating cellular functions with the large body of data thus generated has so far been difficult, mainly because there are no corresponding quantitative measures of cellular functions to parallel the gene expression data. Knowing which genes or groups of genes have similar expression patterns is interesting, but such knowledge would be much more useful if it could be correlated with simultaneous measurements of the many metabolic processes that these expressed genes may regulate. If these data can be obtained, they could potentially unearth many as-yet-unrecognized roles for genes whose functions are incompletely understood, and illuminate the feedback mechanisms and pathways responsible for controlling their expression levels. To this end, we propose to develop multidimensional Nuclear Magnetic Resonance (NMR) and associated computational techniques that will detect the level of as many metabolites as possible in vivo in a high-throughput fashion. Because the small molecules in a cell are a direct and quantitative characterization of its metabolic functions, the data obtained with our | | |

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| **Task Description:** | method will complement the DNA micro-array data and assist efforts to interpret them.<br>The project has six specific aims, encompassing both experimental and modeling components: (i) Developing 13C-1H NMR techniques for detecting small molecule metabolites by suppressing macromolecule signals; (ii) Building a spectral library of small-molecule metabolites, which will be used to develop deconvolution algorithms to automatically assign the NMR spectra of complex mixtures; (iii) Developing experimental and algorithmic methods of identifying compounds from NMR spectra of complex mixtures using NMR spectra measured under different conditions (without using the spectral library); (iv) Investigating the metabolic profile differences between thermophilic and mesophilic bacteria; (v) Implementing computational algorithms that apply Hill's generalized cycle kinetic theory to interpret the metabolite data and (vi) Deriving flux control coefficients using transient metabolite concentrations.<br><br>Developing these technologies will help NASA to achieve its missions in astrobiology, specifically in understanding the origin of intermediary metabolism, and inventing novel biomarkers for detecting life elsewhere. Additionally, understanding the wisdom of billions of years of evolution that have been distilled in the design of metabolic control networks that are robust and failure-tolerant may lead to new biology-inspired design principles for developing highly reliable deep space probes. We also expect practical applications of NMR metabolite profiling technologies to bioengineering and understanding drug degradation. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | No progress report this peirod. |
| **Bibliography Type:** | Description: (Last Updated: 02/27/2004) |